



Research paper

The effects of weather, harvest frequency, and rotation number on yield of short rotation coppice willow over 10 years in northern Japan

Hisanori Harayama^{a,1,*}, Akira Uemura^{a,1}, Hajime Utsugi^b, Qingmin Han^{a,1}, Mitsutoshi Kitao^a, Yutaka Maruyama^c^a Hokkaido Research Center, Forestry and Forest Products Research Institute (FFPRI), 7 Hitsujigaoka, Toyohira, Sapporo, 062-8516, Hokkaido, Japan^b Principal Research Director, FFPRI, 1 Matsunosato, Tsukuba, Ibaraki, 305-8687, Japan^c Department of Forest Science and Resources, Nihon University, 1866 Kameino, Fujisawa, Kanagawa, 252-0880, Japan

ARTICLE INFO

Keywords:

Air temperature
Harvest interval
Hours of sunshine
Precipitation
Salix spp
Woody biomass

ABSTRACT

Under global warming, expectations for woody biomass as a renewable energy are increasing. Short rotation coppice willow is the most widely planted energy wood crop. In this study, we examined the effect of harvest frequency on willow yield across multiple harvest rotations, something which is poorly understood. We planted 18 clones of *Salix pet-susu* and *S. sachalinensis* at a density of 25,000 or 16,667 cuttings ha⁻¹ in Hokkaido, northern Japan, and harvested either annually or biennially for 10 years (2009–2018). Annual variation in willow yield per stool was analyzed using five linear mixed-effect models incorporating the number of rotations, harvest interval, and weather parameters during the growing season with clones and plant density as random factors. Number of rotations had a significant negative effect on annual yield in all five models. We estimated that yield could be halved over 10 consecutive years of harvesting. In three out of the five models, biennial harvesting had a significant positive effect on yield in comparison with annual harvesting. In addition, we found that weather conditions, such as hours of sunshine during growing season, air temperature in May (when leaf flushing occurs), and precipitation in July (when vigorous height growth occurs) have significant positive effects on willow yield. These results suggest that, to maintain willow vigor for longer time periods biennial, or more than two-year-harvest, cycle is recommended rather than annual harvest.

1. Introduction

Emissions of CO₂ and other greenhouse gases due to human activities have been increasing every year, and global warming has been progressing unabated for the past 60 years, already impacting people, ecosystems, and livelihoods globally [1]. In order to minimize global warming to 1.5 °C above pre-industrial levels, which will prevent irreversible loss of ecosystems and crises for people and societies, unprecedented transitions are necessary in all aspects of society, including converting from fossil fuels to renewable energy [1,2]. Wood biomass is a promising resource [3], with short rotation coppice (SRC) willows being one of the most widely planted energy wood crops commercially,

especially in the cool climates of Sweden, the United Kingdom, Ireland, Denmark, and Canada [4–6]. Other European countries, the United States, and New Zealand have also attempted to introduce SRC willows [5–9]. In Japan, the use of woody biomass for energy has been increasingly supported by the feed-in tariff scheme after the Great East Japan Earthquake in 2011 [10]. However, current biomass materials mainly consist of domestic lumbering and building material waste, thinning wood, forest scraps [10] and imported palm kernel shells [11]. There is currently no commercial SRC willow in Japan, but there have been some experimental trials [12–15]. This is despite the fact that many parts of Japan have approximately 1000 mm or more of precipitation during the growing season, which is suitable for SRC willows [16].

Abbreviations: AIC, Akaike's information criterion; a.s.l., above sea level; Dens., initial plant density; Freq., harvest frequency; Int., interaction; LMM, linear mixed-effect model; R^2_{LMM} , conditional coefficient of determination of linear mixed-effect model; Precip., precipitation; Rotat., number of rotations; SRC, short rotation coppice; VIF, variance inflation factor; Sunsh., hours of sunshine; Temp., air temperature.

* Corresponding author. Hokkaido Research Center, Forestry and Forest Products Research Institute (FFPRI), 7 Hitsujigaoka, Toyohira, Sapporo, Hokkaido, 062-8516, Japan.

E-mail address: harahisa@ffpri.affrc.go.jp (H. Harayama).

¹ Present address. Department of Plant Ecology, FFPRI, 1 Matsunosato, Tsukuba, Ibaraki, 305-8687, Japan

<https://doi.org/10.1016/j.biombioe.2020.105797>

Received 8 May 2020; Received in revised form 11 September 2020; Accepted 22 September 2020

Available online 3 October 2020

0961-9534/© 2020 Elsevier Ltd. All rights reserved.

In SRC willow plantations, the willows typically can be harvested and coppiced six to eight times with 2–5 y intervals without replanting, giving the plantation a life span of 19–25 y [17]. However, most studies about SRC willow yield have been conducted only from the first to fourth rotation [18–22], and few studies have evaluated yield throughout the lifespan of plantations. In addition, in willow plantations that are used for the production of cuttings, harvest intervals are on a one-year cycle, because cuttings are normally derived from one-year-old shoots to ensure steady growth and high biomass production [17,23]. However, yields with annual harvest intervals are generally lower than those with longer harvest intervals [21,24–26]. Cuttings from these plantations are needed to establish new SRC willow plantations. The selection of clones is very important to get high yields in SRC willows [27–31], because genetic variations in growth and yield are very large in willows even within species [20,30,32]. If the commercial cultivation of SRC willows is to be introduced and the planting area to be increased each year, a large number of cuttings will need to be harvested every year. This is because of the high planting density needed for SRC willow, normally 15,000 cuttings ha^{-1} [17]. Therefore, it is important to understand how repeated harvests affect the yield of willow biomass. However, our understanding remains inadequate because few studies have conducted repeated harvests [25].

One of the reasons for the uncertainty in the impact of a large number of harvests on biomass yield is the high inter-annual variation in yields of willow [33,34], which are affected in part by weather condition, such as air temperature and precipitation [35,36]. Little has, however, been reported on the quantitative assessment of the effects of weather conditions on the inter-annual variation of willow yield. In this study, we conducted consecutive annual and biennial harvests in an experimental willow field during a ten-year period in northern Japan. We evaluated the effect of the number of harvests together with weather conditions on willow yield per stool using statistical models. Our main objectives were to clarify 1) whether repeated harvests have a negative impact on willow yield per stool and 2) when and what weather conditions affect annual willow yield per stool.

2. Materials and methods

2.1. Study site and planting design

The study site was established in Shimokawa town, northern Hokkaido, Japan (44°15' N, 142°38' E, 200 m a.s.l.). On average, the warmest month is August when daily maximum temperatures often exceed 25 °C, and the coldest month is January when daily minimum temperatures are often less than –10 °C. The site was 33 m × 12 m and almost flat. The soil was a silty clay from 0 to 11 cm depth and a sandy loam from 11 to 30 cm depth. All cuttings were planted by hand in

November 2007 and were cut back (coppiced) in November 2008 to promote sprouting according to typical cultivation methods [37]. A slow release chemical fertilizer (N:P:K = 12:8:5, Shin-maruyama tokugou, Nihon ringyo hiryou) was applied at a rate of 20 g m^{-2} , i.e., 24 kg N ha^{-1} , 16 kg P ha^{-1} , and 10 kg K ha^{-1} , every May after the snow thaw from 2008 to 2018.

Cuttings with the recommended size in the general protocol [17] (ca. 20 cm long and ca. 1 cm diameter) of 12 different clones of *Salix pet-susu* Kimura and 6 different clones of *S. sachalinensis* F.Schmidt were planted at two planting densities—either in triple and double row systems (Fig. 1). In a 3 × 4 m plot, 15 cuttings of a single clone were planted in half of the plot (subplot) in three rows at a density of 25,000 cuttings ha^{-1} with 0.5 m between rows and 0.5 m between cuttings within rows. Ten cuttings of the clone were planted in the other half of the plot in 2 rows at a density of 16,667 cuttings ha^{-1} with 1.0 m between rows and 0.5 m between cuttings within rows (Fig. 1A). Experimental blocks with biennial and annual harvest intervals were constructed with 16 plots planted with different clones, each plot constituting of two subplots with the two planting densities described above, respectively (Fig. 1B). The biennial and annual harvest blocks were, therefore, composed of the same design for planting density. Fourteen of the 18 clones were planted in both blocks. The remaining four clones were planted in only one block because the number of cutting was insufficient. Specifically, the clones P.G.A and P.G.B were planted in the biennial harvest block, and P.G.C and P.G.E in the annual harvest block. Note that willow clones in this study did not promise high yield, because clone selection for SRC willow is still underway in Japan.

2.2. Harvesting and measurement

At the end of growing seasons (late October to early November) from 2009 to 2018, willows were harvested manually with a saw and pruning scissors at approximately 15 cm from the ground according to the harvest interval designs. Fresh biomass of aboveground organs per stool were weighed immediately from each three to five cuttings per subplot in each plot, with the exception of clone P.I.82 in the annual harvest block and P.G.12 in the biennial harvest block in 2018, which were weighed from only two cuttings in 16,667 ha^{-1} subplots due to the low survival in P.I.82 and due to mis-operation in P.G.12. To obtain oven-dry mass yield, representative subsamples of the fresh harvests were weighed in each plot, and taken back to the laboratory, dried at 70 °C until they reached a constant mass, and reweighed. The oven-dry mass yield per stool was calculated by multiplying the fresh weight by the ratio of dry to fresh weight of the subsample from each plot. The biomass yields from the biennial harvest block were divided by two to convert to annual yields. Note that the yield data for the annual harvest of P.G.C and P.G.E in 2009 and the biennial harvests for all clones in 2012 were

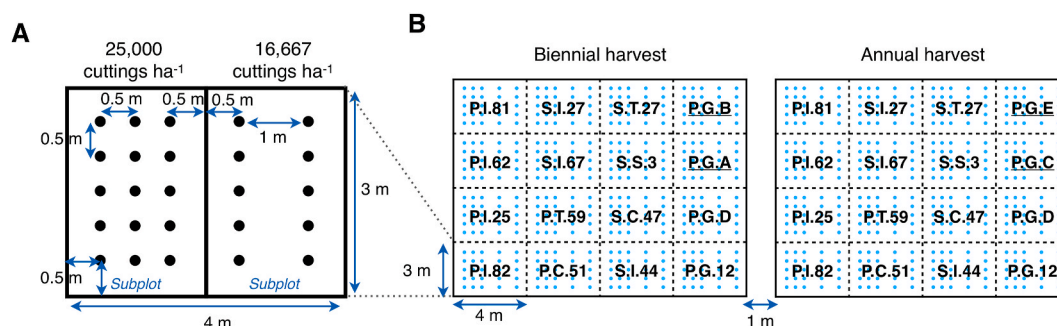


Fig. 1. Experimental layout: (A) planting design with two planting densities in a plot; (B) clone arrangement in a block with biennial and annual harvest intervals. Clone names starting with “P” represent *Salix pet-susu*, and those with “S” represent *S. sachalinensis*. The light blue dots in panel B represent the planting points in each plot. Fourteen of the 18 clones were planted in both blocks, but four clones (underlined) were planted in only one block (P.G.A and P.G.B in the biennial harvest block and P.G.C and P.G.E in the annual harvest block). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

not available due to data storage problems. In November 2018, when the last yield measurement was made, we counted the number of surviving and dead stools in all subplots, and calculated last plant density and survival rate for each subplot.

2.3. Data analysis

All statistical analyses were performed with R version 3.6.2 [38]. Effects of harvest frequency (annual or biennial), number of rotations (1–10), and weather parameters during the growing season (May to October) (Fig. 2) on oven-dry mass yield per stool were explored using linear mixed-effect models (LMM) with a log-link function by restricting the maximum likelihood using the *lmer* function from the R package “lme4”. As weather data, we analyzed air temperature, precipitation, and duration of sunshine during the growing seasons from 2009 to 2018, which obtained at the nearest weather station (44°18' N, 142°38' E, 143 m a.s.l., Shimokawa weather station, Japan Meteorological Agency). An LMM simultaneously incorporated monthly temperature, precipitation, and duration of sunshine data could not be analyzed due to multicollinearity among variables. We, therefore, constructed four models incorporating each four sets of weather data as fixed effects: (1) annual data of mean air temperature, precipitation, and duration of sunshine (Fig. 2A,B,C) was used in the annual weather model, (2) monthly data of mean air temperature (Fig. 2D) was used in the monthly temperature model, (3) monthly data of precipitation (Fig. 2E) was used in the monthly precipitation model, and (4) monthly data of duration of sunshine (Fig. 2F) was used in the monthly sunshine model. Additionally, we constructed a model which incorporated all significant variables (p

< 0.05) from the above four models as fixed effects and then selected the lowest Akaike's information criterion (AIC) using the *dredge* function from the R package “MuMIn”. Since the difference in AIC between the top two models was less than 2, we determined the best model with the highest conditional coefficients of determination (R^2_{LMM}) calculated using the *r.squaredGLMM* function from the R package “MuMIn”, we called this the combined weather model. We added harvest frequency (annual or biennial) and number of rotations to the fixed effects in each model, in addition to the weather parameters described above, because they were expected to affect the annual variation in yield. Since we determined annual harvest as the reference category for harvest frequency, the effect of harvest frequency on yield was analyzed for the effect of biennial harvest compared to annual harvest. We focused on the overall effects of harvest frequency, harvest cycle and weather condition on yield in this study, we, thus, incorporated other parameters that were expected to affect yield, i.e., initial and final plant densities and clones, as random factors. In constructing models, we normalized data of harvest time and weather parameters, as well as oven-dry mass yield per stool, so that the mean was 0 and the standard deviation was 1 using the *scale* function in the base R program in order to compare the fixed effects among the variables. We calculated the variance inflation factor (VIF) of each model that contained more than one weather parameter, i.e., the annual weather model and the combined weather model, to evaluate multicollinearity among variables in each month ($\text{VIF} > 10$). Since we found multicollinearity among air temperature, precipitation, and hours of sunshine in May in constructing the combined weather model, we constructed three models excluding one of each of the three weather factors, and we used the model with the lowest AIC, i.e. the model

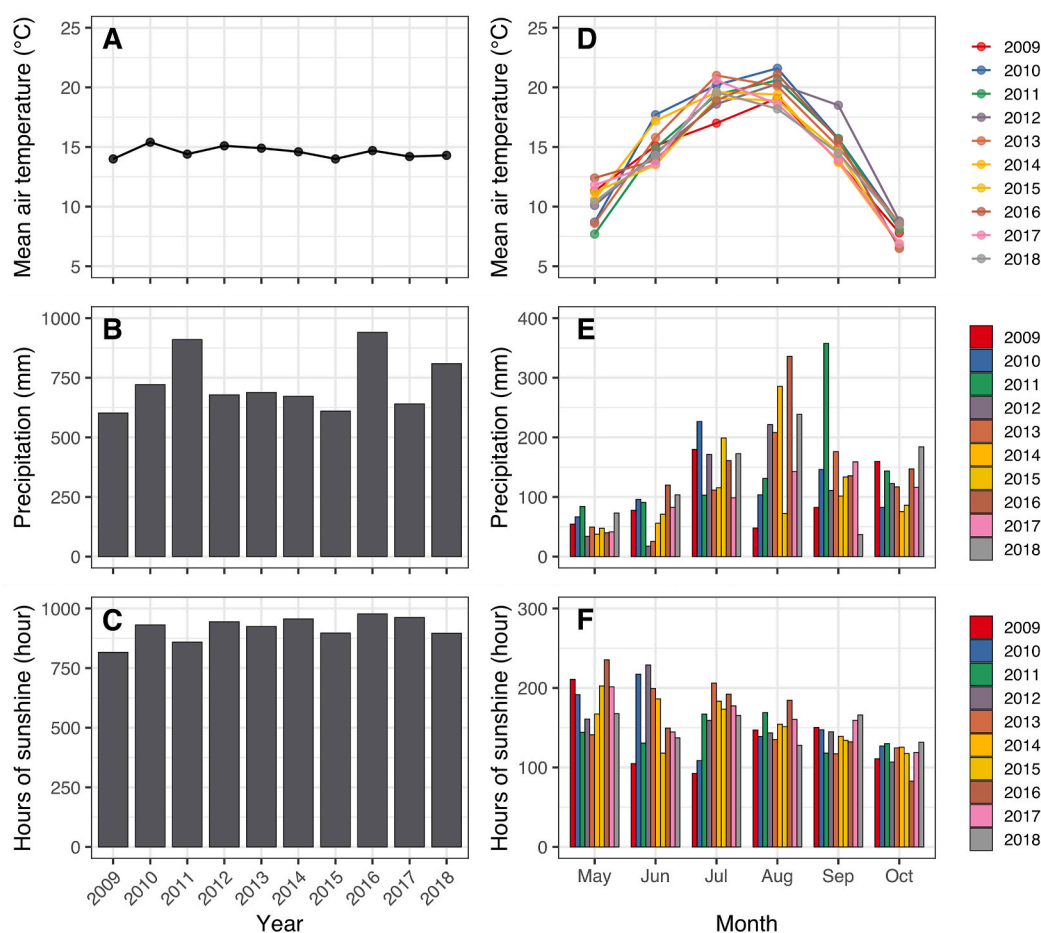


Fig. 2. Annual and monthly weather data for mean air temperature (A, D), precipitation (B, E), and hours of sunshine (C, F) during the growing season (May 1 to October 31) from 2009 to 2018.

excluding hours of sunshine in May, for the subsequent model selection (the AIC of the model excluding hours of sunshine, precipitation, and air temperature was 3492, 3497, and 3499, respectively). We compared the AIC and R^2_{LMM} of the five models to rank the models in terms of simplicity and predictive power, determining the model with the lowest AIC and highest R^2_{LMM} as the best model. The best model was re-modeled using a non-normalized dataset to determine the effect of significant variables on dry mass yield per stool with the *lmer* function and the *plot_model* function from the R package “sjPlot”.

Effect of harvest frequency and initial plant density on survival rate and final plant density 10 years after first harvest were analyzed generalized linear mixed models assuming a binomial distribution for survival rate and a gaussian distribution for final plant density, using *lmer* and *glmer* function from the R package “lme4”. Clones were incorporated into the random factors. A level of significance of $p = 0.05$ was used to determine the statistical significance.

3. Results

3.1. Weather data from the growing seasons during the study period

During the growing seasons of the study period, annual mean air temperature was 14–15 °C with little annual variation (Fig. 2A). Annual precipitation was variable from year to year, ranging from 610 mm in 2009 to 940 mm in 2016 (Fig. 2B), whereas annual hours of sunshine was relatively less variable, ranging from 816 h in 2009 to 977 h in 2016 (Fig. 2C). Monthly mean air temperature was low in May and October, at around 10 and 8 °C, and high in July and August, at around 20 °C, and the degree of the annual variation in monthly mean temperature was generally similar in each month (Fig. 2D). Monthly precipitation tended to be low in May and July and high in August and September, and the annual variation in monthly precipitation tended to be high in August and September and low in May and October (Fig. 2E). Hours of sunshine tended to be high in May and low in October, and annual variation of

hours of sunshine tended to be high in May to July (Fig. 2F). Overall, monthly weather data tended to have more variation than the annual weather data.

3.2. Annual and biennial variations in yields of willow and survival rate

Oven-dry biomass yields of willow from annual harvests tended to be more variable from year to year compared with those from biennial harvests, and tended to be highly variable among clones (Fig. 3). In some clones (e.g. P.G.C and P.I.81), yields from the annual harvest block tended to be higher in earlier rotations and then decline as the number of rotations increased, although there were some exceptional years. In some clones (e.g. S.I.27 and S.T.27), yield from the annual harvest block tended to be high up until the third or fourth rotation and then remain low for later rotations. In other clones (e.g. P.G.12, P.T.59, and S.I.67), there were no clear trends in yield with annual harvesting. In the biennial harvest block, there were no consistent trends in yield over time among clones. In some clones, yields peaked in 2010, 2012, and 2016.

Survival rate 10 years after first harvest was varied from 0% to 100% among clones, harvest frequency and initial plant densities (Fig. 4A). There was no significant effect of harvest frequency or initial plant density on survival rate ($p > 0.05$). Final plant density was also varied from 0 to 25000 plant per ha⁻¹ (Fig. 4B). Initial plant density had a significant effect on final plant density, whereas harvest frequency had no significant effect on it.

3.3. Effects of weather and consecutive harvests on annual willow yields analyzed by LMM

Annual biomass yields of willow were significantly affected by weather conditions such as air temperature, precipitation, and hours of sunshine during the growing season, as well as harvest frequency and number of rotations (Fig. 5). In the annual weather model, annual yields were most strongly affected by the number of rotations. The effect of

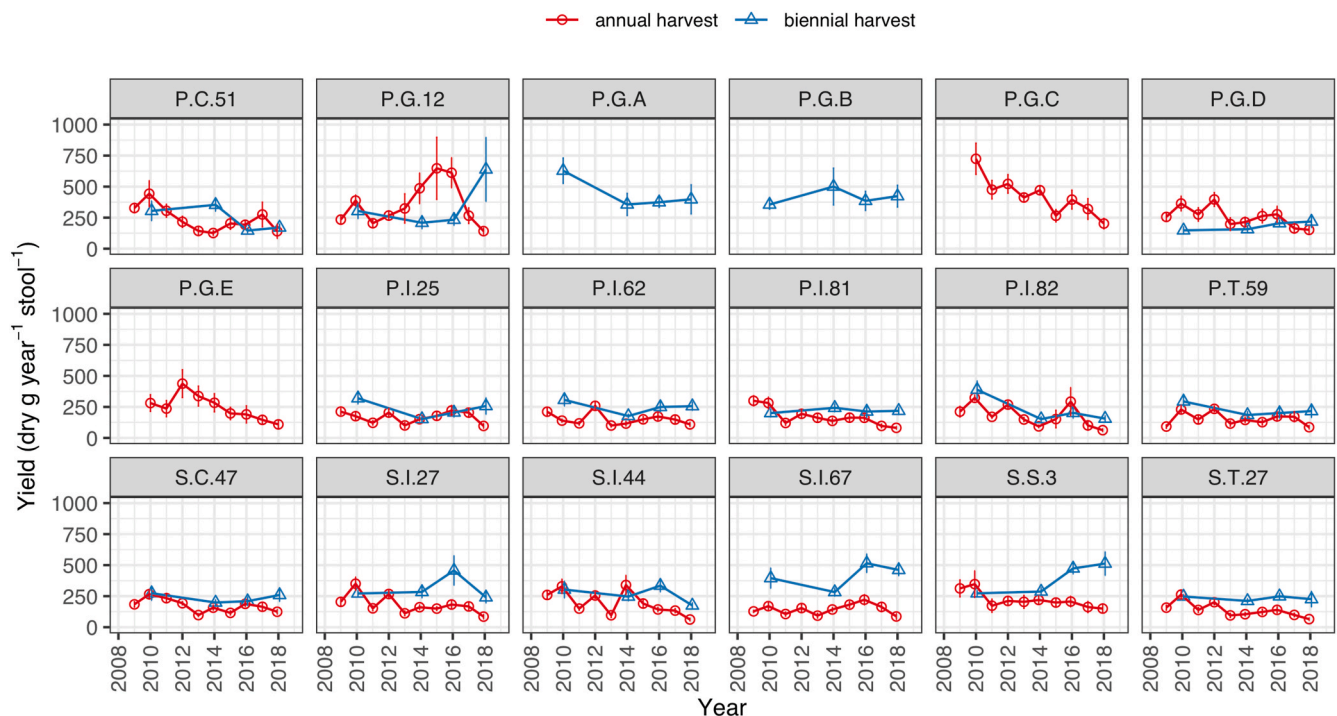


Fig. 3. Annual variations in yield for each of the 16 willow clones with annual and biennial harvests over a 10-year period. Clone names starting with “P” represent *Salix pet-susu*, and those with “S” represent *S. sachalinensis*. Note that in 2012, we harvested from both annual and biennial harvest blocks, but the data from the biennial block are not available due to an accident. Error bars represent standard errors of the mean ($n = 6$ –10 with exceptions of $n = 2$ in P.I.82 for the annual harvest and P.G.12 for the biennial harvest in 2018).

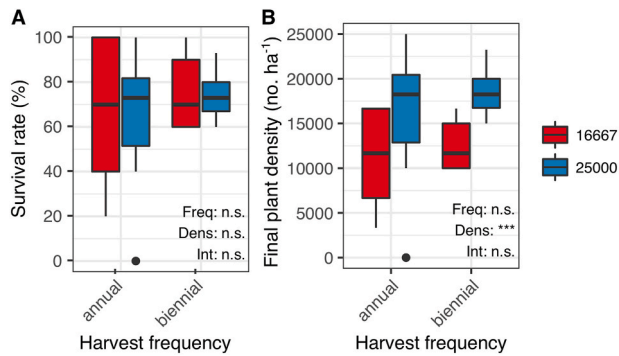


Fig. 4. Comparison between harvest frequency and initial plant density of survival rate (A) and final plant density (B) 10 years after the first harvest. The values of the subplot in each clone were pooled. Results of generalized linear mixed model were added. Freq; harvest frequency, Dens; initial plant density, Int: interaction. n.s.; $p > 0.05$, ***; $p < 0.001$.

number of harvests was significant and negative. Annual yields were also significantly positively affected by harvest frequency (biennial harvest relative to annual harvest) and hours of sunshine during the growing season. However, annual yields were not significantly affected by precipitation or air temperature (Fig. 5A). In the other four models, the number of rotations also had the strongest effect on annual yield, with a significant negative effect in all models. In the monthly temperature model, air temperature in May had the greatest significant positive effect on annual yield, and harvest frequency (biennial harvest relative to annual harvest) and air temperatures in September and July also had significant positive effects (Fig. 5B). In the monthly precipitation model, harvest frequency (biennial harvest relative to annual harvest) and precipitation in July and August had significant positive effects and precipitation in May had a significant negative effect on annual yield (Fig. 5C). In the monthly sunshine model, hours of sunshine in May had a significant positive effect on annual yield (Fig. 5D). In the combined weather model, which incorporated all significant annual and monthly weather parameters from the four models and then was simplified on the

basis of AIC, annual hours of sunshine, air temperature in May, and precipitation in May and July were significant fixed effects on annual yield (Fig. 5E). Annual hours of sunshine during the growing season had the greatest positive effect on annual yield, and air temperature in May and precipitation in May and July also had significant positive effects on annual yield.

The models incorporating monthly weather data, i.e. the monthly temperature, precipitation, and sunshine models, had lower AIC and higher R^2_{LMM} than the annual weather model (Table 1). The combined weather model had the lowest AIC and highest R^2_{LMM} among the five models, and thus was determined to be the best model, whereas the R^2_{LMM} (0.354) was almost the same as the monthly models (0.333–0.349), which had moderate predictive power.

Scatter plots with the trends between each variables of fixed-effects and predicted yields reanalyzed by incorporating the non-normalized data set into the combined weather model are shown in Fig. 6. It was estimated that the annual yield of willow significantly decreased by $18.16 \pm 1.84 \text{ g year}^{-1} \text{ stool}^{-1}$ per rotation, resulting in a yield loss of $181.6 \text{ g year}^{-1} \text{ stool}^{-1}$ after 10 harvests (Fig. 6A). Based on the estimates of each of the other variables, it was also estimated that 100 more hours of sunshine during the growing season could significantly increase annual yields by $74.9 \text{ g year}^{-1} \text{ stool}^{-1}$ (Fig. 6B), an increase in temperature of 5°C in May could significantly increase annual yields by $72.2 \text{ g year}^{-1} \text{ stool}^{-1}$ (Fig. 6C), and an increase in precipitation of 50 mm in May and July could significantly increase annual yields by 54.6

Table 1

Akaike's information criterion (AIC) and conditional coefficients of determination (R^2_{LMM}) for five linear mixed-effect models (with clones and plant density as random factors) predicting annual variation in dry yield of willow per stool. Each model was fitted to 1374 observations of dry yield per stool.

Model	df	AIC	R^2_{LMM}
Annual weather model	10	3510	0.333
Monthly temperature model	13	3506	0.346
Monthly precipitation model	13	3505	0.348
Monthly sunshine model	13	3500	0.349
Combined weather model	10	3472	0.354

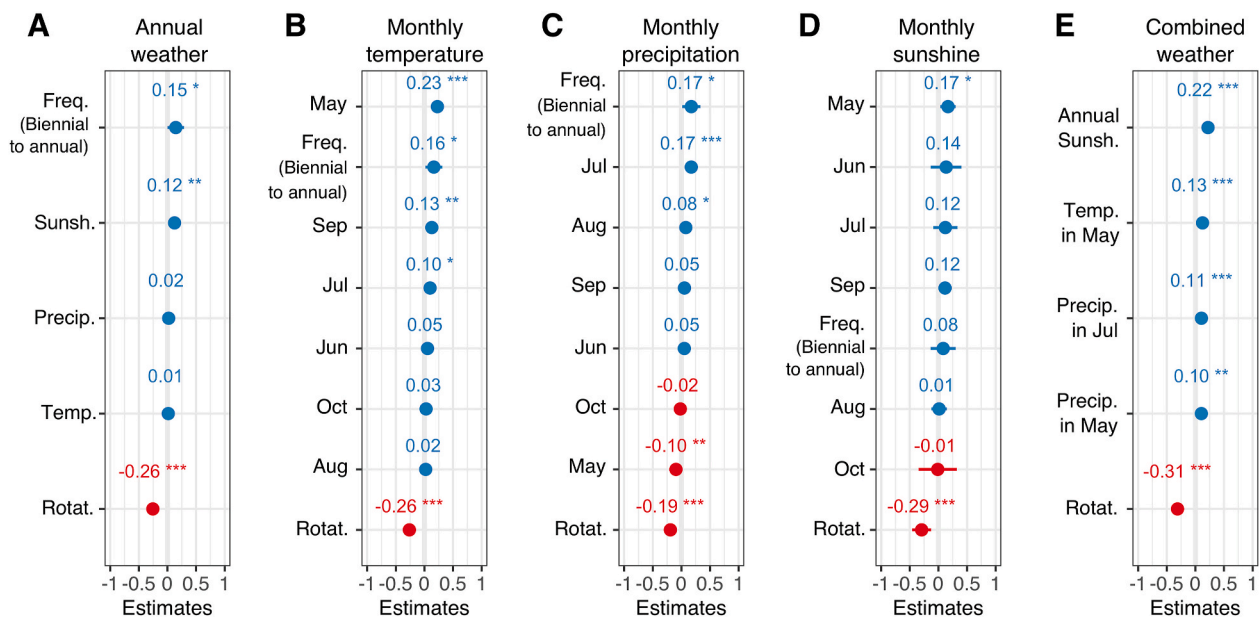


Fig. 5. Estimates of fixed effects from five linear mixed-effect models (with clones and initial and final plant densities as random factors) predicting annual variation in dry yield of willow per stool. (A) annual weather model, (B) monthly temperature model, (C) monthly precipitation model, (D) monthly sunshine model, and (E) combined weather model. Each model was fitted to 1374 observations of dry yield per stool and all numerical variables were normalized. Freq.; harvest frequency, Sunsh.; hours of sunshine, Precip.; precipitation, Temp.; air temperature, Rotat.; number of rotations. Error bars represent 95% confidence intervals. *, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$.

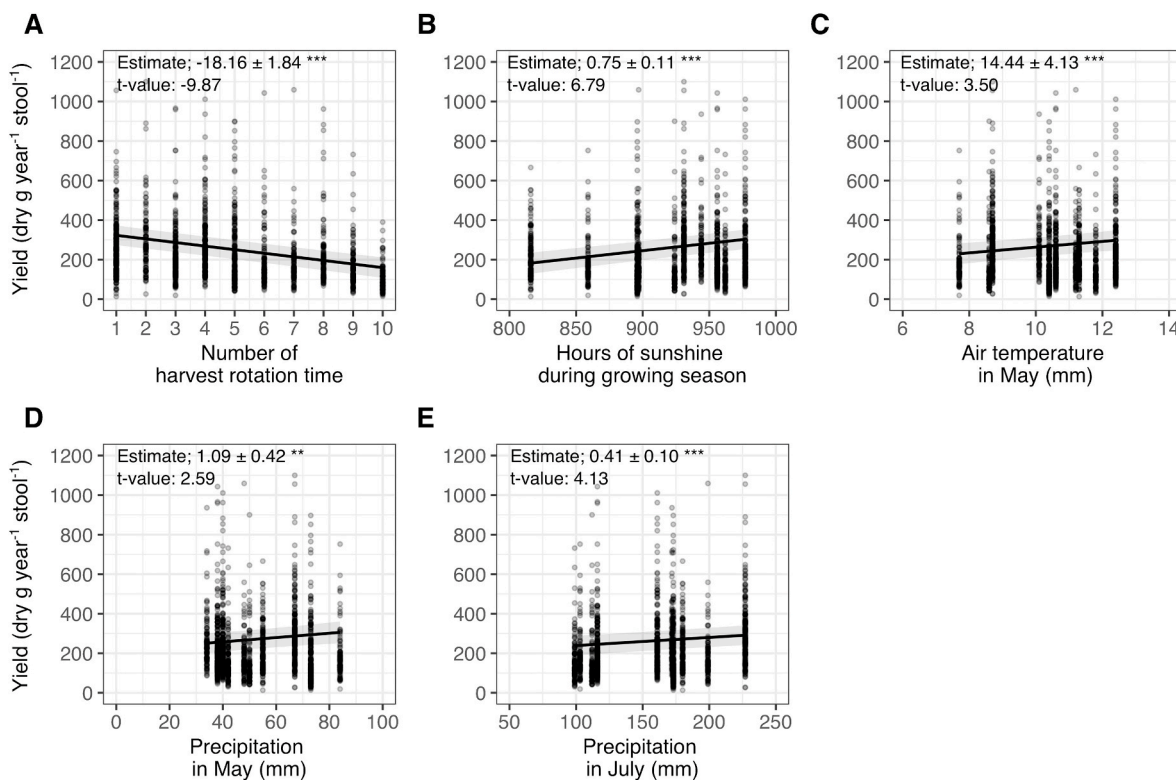


Fig. 6. Estimated trends of fixed-effects of (A) number of rotations, (B) hours of sunshine during the growing season, (C) air temperature in May, (D) precipitation in May, and (E) precipitation in July on annual yield of willow predicted by a linear mixed-effect model based on the combined weather model using non-normalized data sets ($n = 1374$). The black lines represent the predicted trends and the shaded areas represent 95% confidence intervals of the predicted line. Estimate values, standard errors, and t-values of the fixed effects are described. ***, $p < 0.001$, **, $p < 0.01$.

and 20.6 g year⁻¹ stool⁻¹, respectively (Fig. 6D and E).

Evaluating the random factors of the combined weather model, the yield per stool tended to be higher in clones such as P.G.C, P.G.A, P.G.12, P.G.B, P.G.E, and S.S.3, and lower in clones such as S.T.27, P.T.59, P.I.82, P.I.81, and P.I.62 (Fig. 7A). The lower initial plant density (16667 plants ha⁻¹) tended to be have higher yield per stool than the higher density (25000 plants ha⁻¹) (Fig. 7B), and the final plant density also

tend to have higher yields at the lower density, with a few exceptions (Fig. 7C).

4. Discussion

Although the yield was highly variable among clones and among years over the ten-year experimental period (Fig. 3), we were able to

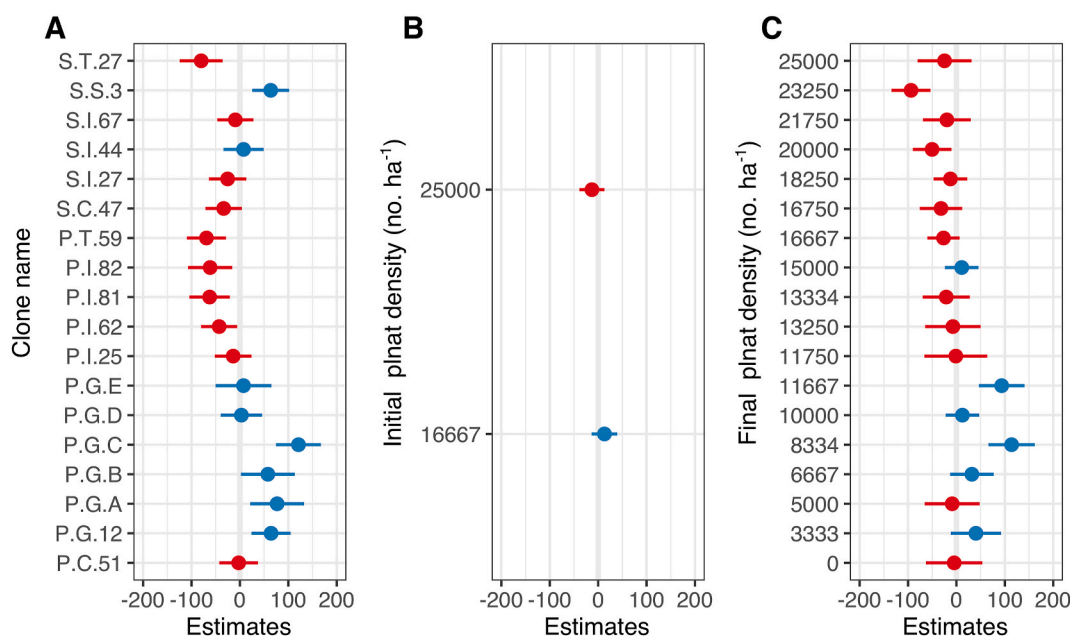


Fig. 7. Estimates of random factors of combined weather models using non-normalized data sets ($n = 1374$). Error bars 95% confidence intervals.

clearly elucidate the negative effect of number of rotations (up to 10) on annual willow yield per stool using an LMM approach incorporating weather parameters (Fig. 5). A similar downward trend in willow yield with increasing number of rotations was reported for five willow cultivars planted at four densities in twelve consecutive annual harvest rotations by Stolarski et al. [25]. However, the unit of yield was not per stool but per planted area so that the decreasing yield partly resulted from a reduced survival rate during the experimental period, and analyzed with more focus on the survival rate [25]. Our results demonstrated that the reduction in willow yield due to repeated harvesting can be also caused by lower yields per stool. On the other hand, in the previous paper by Stolarski et al. in which they analyzed data for the first ten of the twelve years [34], they did not find a negative effect of consecutive annual harvest on yield and did not provide a clear interpretation of the fluctuated trend in yields other than the effect of precipitation. In a series of studies by Stolarski et al. [25,34], the effect of harvest frequency on yield was unclear with low yielding planting densities and clones, whereas the negative effect of harvest frequency was pronounced with high yielding planting densities and clones. Our study has also found that low yield clones, such as S.T.27 and P.T.59, tended to obscure trend of yield reduction with increasing harvest frequency. These results suggest that although increasing the number of rotation generally can have a negative effect on yield, the effect of rotation number can be masked by other limiting factors that result in low yields, for example potentially low yield clones and high plant density.

The yield loss per stool per rotation estimated from the best model in this study (Fig. 6A) was equivalent to approximately 5% of the initial average yield. Thus, yield could be halved in 10 consecutive years of harvesting. The reduction in willow yield per stool with increasing number of harvests could be due to damage to the stool at harvesting [39], weed overgrowth due to inadequate willow canopy closure [40], reduced soil nutrients due to insufficient fertilizer application [41], and a reduction of carbohydrate reserves in the roots and cut stump [42]. In addition, it was estimated that biennial harvesting had a significant positive effect on willow yield in comparison to annual harvesting in three out of the five models constructed (Fig. 5ABC). In fact, the roots of willows from the biennial harvest block tended to be bigger than those from the annual harvest block (Fig. S1). Although plant density can affect root growth, differences in harvest frequency would be a direct cause of the difference of root size, because there was no significant difference in survival rate of final plant density in harvest frequency (Fig. 4). Since annual harvest possibly can result in lower carbohydrate accumulation in the roots and lower water and nutrient absorption in willows in the annual harvest block, it might be preferable to harvest biennially, or even less often, rather than annually in order to keep the stool vigorous over a decade without yield declines per stool. This could be particularly important in regions where willow SRC breeding is limited, because it is not possible to replant new, better clones with the promise of higher yields. It should be noted, however, that in the plantations for production of cuttings (as opposed to biomass production), biennial harvesting may increase costs as cuttings from one-year-old shoots are preferable to those from two-year-old cuttings [17].

Willow yield per stool was also affected by annual and monthly weather conditions (Fig. 5). Among the weather parameters measured, the air temperature in May and the annual duration of sunshine had the strongest influence on yield per stool. The large positive effect of annual duration of sunshine reflects the ecological traits of willow—it is a pioneer and light-demanding species. In addition, the large positive effect of air temperature in May could be because warmer temperatures in May promote early leaf flushing which increases the growing period of willow [8,33,43]. July precipitation also had a large effect on willow yield. Willow species are one of the most vulnerable species to xylem cavitation among temperate tree species [44]; that is, willow species are very susceptible to xylem dysfunction in drying soils. Studies on SRC

willow in the United Kingdom found that rainfall and soil moisture dominate in determining regional willow yield in the UK [19,36]. Hence, precipitation in July, which is when willow shows vigorous height growth in northern Japan, could have a strong effect on willow yield.

While we determined the effects of harvest frequency and weather conditions on willow yield, the predictive power of our models was only moderate. Although we incorporated the clones as a random factor in these models (Fig. 7), the annual variation in yield varied widely between clones (Fig. 3), suggesting that there could be an interaction between clones and the number of harvests and/or each weather parameter. In other words, different clones may have different favorable weather conditions and different harvest impacts, but the models in this study could not take this into account. This probably resulted in a reduction in the predictive power of our models. Similar interannual variability in yield among clones has previously been reported in willow cultivars in Europe [34]. The average yield per stool for most clones in this study was low, less than 250 g year⁻¹, corresponding to 3.75 dry tonne ha⁻¹ year⁻¹ at a density of 15,000 cuttings ha⁻¹ even ignoring the post-planting mortality, which far short of the general target yield of 10 dry tonne ha⁻¹ year⁻¹. Therefore, it is necessary to promote the selection and breeding of high yielding willow clones in Japan like those in Europe and the United States [31]. The LMM approach used in this study will also be useful for quantitatively analyzing the annual variation in yields of other willow clones, such as varieties that are widespread in Europe and the United States and high-yield varieties that will be selected in the future in Japan, although trend in the influence of each factor may differ from this study. In the future, it will be important to develop models with higher predictive power by taking into account the interactions between clones, harvest frequency, and weather parameters, as well as other parameters, such as insect outbreaks, diseases, soil, and groundwater, which were not considered in this study. Furthermore, yield of SRC willow plantation do not always reflect per stool, as planting density and survival rate also have a significant impact on the plantation yield [26,45]. Incorporating changes in survival rate over time, which could not be include in this study, into the model will allow for the prediction of yield per area, providing more useful information for predicting profitability of SRC willow plantation.

5. Conclusions

We analyzed annual variation in willow yield over ten years using models that incorporated harvest and weather parameters. We found that repeated annual harvests could have a negative effect on willow yield per stool, potentially halving yields over a 10-year period. Also, our results suggest that annual harvesting could result in lower yields than biennial harvesting. Of the weather conditions measured, the hours of sunshine during the growing season, air temperature in May (when leaf flushing occurs), and precipitation in July (when vigorous height growth occurs) had the strongest effects on yield in the willow clones studied. Further study to analyze the annual variation in yield of other willow clones with potentially high-yielding using the model analysis in this study will be useful for improving SRC willow production, in estimating productive years for willow cuttings and SRC, as well as for predicting willow yield under current and future climate.

Funding

This work was supported by JSPS KAKENHI [grant number JP24580231]; NEDO [grant number P09014]; FFPRI [grant number #200805].

CRedit authorship contribution statement

Hisanori Harayama: Conceptualization, Methodology, Investigation, Writing - original draft, Project administration. **Akira Uemura:**

Conceptualization, Methodology, Investigation, Funding acquisition. **Hajime Utsugi**: Investigation, Project administration, Funding acquisition. **Qingmin Han**: Investigation, Writing - review & editing. **Mitsutoshi Kitao**: Investigation, Writing - review & editing. **Yutaka Maruyama**: Conceptualization, Methodology, Investigation, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgments

We thank collaborators, Shimokawa town, especially Yuji Takahashi and Toshio Yamamoto, for establishment of this trial, maintenance and harvesting, and Oji Holdings Corporation for offering the willow cuttings. We also thank Haruka Yamamoto, Makoto Ishihara, Eriko Ito, Shuhei Aizawa and other staff at the Hokkaido research center of FFPRI for their assistance in data collection.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biombioe.2020.105797>.

References

- [1] IPCC, Global Warming of 1.5°C, in: V. Masson-Delmotte, P. Zhai, H.-O. Pörtner, D. Roberts, J. Skea, P.R. Shukla, A. Pirani, W.C. Péan, R. Pidcock, S. Connors, J.B. R. Matthews, Y. Chen, X. Zhou, M.I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, T. Waterfield (Eds.), An IPCC Special Report on the Impacts of Global Warming of 1.5°C above Pre-industrial Levels and Related Global Greenhouse Gas Emission Pathways, in the Context of Strengthening the Global Response to the Threat of Climate Change, Sustainable Development, and Efforts to Eradicate Poverty, 2018.
- [2] IRENA, Climate Change and Renewable Energy, National Policies and the Role of Communities, Cities and Regions (Report to the G20 Climate Sustainability Working Group (CSWG)), International Renewable Energy Agency, Abu Dhabi, 2019.
- [3] P. Lauri, P. Havlík, G. Kindermann, N. Forsell, H. Böttcher, M. Obersteiner, Woody biomass energy potential in 2050, *Energy Pol.* 66 (2014) 19–31.
- [4] A. Don, B. Osborne, A. Hastings, U. Skiba, M.S. Carter, J. Drewer, H. Flessa, A. Freibauer, N. Hyvönen, M.B. Jones, G.J. Lanigan, Ü. Mander, A. Monti, S. N. Djomo, J. Valentine, K. Walter, W. Zegada-Lizarazu, T. Zenone, Land-use change to bioenergy production in Europe: implications for the greenhouse gas balance and soil carbon, *GCB Bioenergy* 4 (4) (2012) 372–391.
- [5] K.N. Lindegaard, P.W. Adams, M. Holley, A. Lamley, A. Henriksson, S. Larsson, H. G. von Engelbrechten, G. Esteban Lopez, M. Pisarek, Short rotation plantations policy history in Europe: lessons from the past and recommendations for the future, *Food Energy Secur* 5 (3) (2016) 125–152.
- [6] P.A. Townsend, N. Haider, L.A. Boby, J.P. Heavey, T.A. Miller, A Roadmap for Poplar and Willow to Provide Environmental Services and to Build the Bioeconomy, Washington State University, 2018, p. 37.
- [7] K. Snowden, I. McIvor, I. Nicholas, Short rotation coppice with willow in New Zealand, proceedings of international energy agency bioenergy: short rotation crops for bioenergy, 2013, pp. 1–39.
- [8] B.Y. Amichev, R.D. Hangs, S.M. Konecni, C.N. Stadnyk, T.A. Volk, N. Bélanger, V. Vujanovic, J.J. Schoenau, J. Moukoui, K.C.J. Van Rees, Willow short-rotation production systems in Canada and northern United States: a review, *Soil Science Society of America Journal* 78 (S1) (2014) S168–S182.
- [9] M. Castano-Díaz, M. Barrio-Anta, E. Afif-Khouri, A. Camara-Obregon, Willow short rotation coppice trial in a former mining area in northern Spain: effects of clone, fertilization and planting density on yield after five years, *Forests* 9 (3) (2018).
- [10] Forestry Agency Japan, State of Japan's forests and forest management –3rd country report of Japan to the Montreal Process–, in: Forestry Agency Japan, 2019, p. 153.
- [11] N.A. Pambudi, K. Itaoka, A. Chapman, N.D. Hoa, N. Yamakawa, Biomass energy in Japan: current status and future potential, *international journal of smart grid and clean energy* 6 (2) (2017) 119–126.
- [12] Y. Mitsui, S. Seto, M. Nishio, K. Minato, K. Ishizawa, S. Satoh, Willow clones with high biomass yield in short rotation coppice in the southern region of Tohoku district (Japan), *Biomass Bioenergy* 34 (4) (2010) 467–473.
- [13] T. Yoshioka, K. Inoue, B. Hartsough, Cost and greenhouse gas (GHG) emission analysis of a growing, harvesting, and utilizing system for willow trees aimed at short rotation forestry (SRF) in Japan, *Journal of the Japan Institute of Energy* 94 (6) (2015) 576–581.
- [14] Q. Han, H. Harayama, A. Uemura, E. Ito, H. Utsugi, The effect of the planting depth of cuttings on biomass of short rotation willow, *Journal of Forest Research* 22 (2) (2017) 131–134.
- [15] Q. Han, H. Harayama, A. Uemura, E. Ito, H. Utsugi, M. Kitao, Y. Maruyama, High biomass productivity of short-rotation willow plantation in boreal Hokkaido achieved by mulching and cutback, *Forests* 11 (5) (2020).
- [16] J. Wickham, B. Rice, J. Finnan, R. McConnon, A Review of Past and Current Research on Short Rotation Coppice in Ireland and Abroad, COFORD, Dublin, Ireland, 2010.
- [17] B. Caslin, J. Finnan, C. Johnston, A. McCracken, L. Walsh, Short rotation coppice willow best practice guidelines, in: Teagasc and Agri-Food and Biosciences Institute, 2015.
- [18] M. Stolarski, S. Szczukowski, J. Tworowski, A. Klasa, Productivity of seven clones of willow coppice in annual and quadrennial cutting cycles, *Biomass Bioenergy* 32 (12) (2008) 1227–1234.
- [19] M.J. Tallis, E. Casella, P.A. Henshall, M.J. Aylott, T.J. Randle, J.L.L. Morison, G. Taylor, Development and evaluation of ForestGrowth-SRC a process-based model for short rotation coppice yield and spatial supply reveals poplar uses water more efficiently than willow, *GCB Bioenergy* 5 (1) (2013) 53–66.
- [20] S.U. Larsen, U. Jørgensen, P.E. Lærke, Willow yield is highly dependent on clone and site, *BioEnergy Research* 7 (4) (2014) 1280–1292.
- [21] R.F. Kopp, L.P. Abrahamson, E.H. White, K.F. Burns, C.A. Nowak, Cutting cycle and spacing effects on biomass production by a willow clone in New York, *Biomass Bioenergy* 12 (5) (1997) 313–319.
- [22] M.J. Fernández, R. Barro, J. Pérez, P. Ciria, Production and composition of biomass from short rotation coppice in marginal land: a 9-year study, *Biomass Bioenergy* 134 (2020).
- [23] T. Verwijst, A. Lundkvist, S. Edelfeldt, J. Albertsson, Development of Sustainable Willow Short Rotation Forestry in Northern Europe, in: M.D. Matovic (Ed.), *Biomass Now - Sustainable Growth and Use*, IntechOpen, London, UK, 2013.
- [24] S.U. Larsen, U. Jørgensen, P.E. Lærke, Harvest interval and row spacing of SRC willow influence yield and nutrient content, *Biomass Bioenergy* 126 (2019) 181–189.
- [25] M.J. Stolarski, S. Szczukowski, J. Tworowski, M. Krzyżaniak, D. Załuski, Willow production during 12 consecutive years—the effects of harvest rotation, planting density and cultivar on biomass yield, *GCB Bioenergy* 11 (2019) 635–656.
- [26] S. Szczukowski, M. Stolarski, J. Tworowski, J. Przyborowski, A. Klasa, Productivity of willow coppice plants grown in short rotations, *Plant Soil and Environment* 51(9), 2005, pp. 423–430.
- [27] U. Gullberg, Towards making willows pilot species for coppicing production, *For. Chron.* 69 (6) (1993) 721–726.
- [28] K. Lindegaard, J.H.A. Barker, Breeding willows for biomass, *Aspect Appl. Biol.* 49 (1996) 1–9.
- [29] S. Larsson, Genetic improvement of willow for short-rotation coppice, *Biomass Bioenergy* 15 (1) (1998) 23–26.
- [30] L.B. Smart, T.A. Volk, J. Lin, R.F. Kopp, I.S. Phillips, K.D. Cameron, E.H. White, L. P. Abrahamson, Genetic improvement of shrub willow (*Salix* spp.) crops for bioenergy and environmental applications in the United States, *Unasylva* 56 (2005) 51–55.
- [31] J. Clifton-Brown, A. Harfouche, M.D. Casler, H. Dylan Jones, W.J. Macalpine, D. Murphy-Bokern, L.B. Smart, A. Adler, C. Ashman, D. Awty-Carroll, C. Bastien, S. Bopper, V. Botnari, M. Brancourt-Huvel, Z. Chen, L.V. Clark, S. Cosentino, S. Dalton, C. Davey, O. Dolstra, I. Donnison, R. Flavell, J. Greef, S. Hanley, A. Hastings, M. Hertzberg, T.W. Hsu, L.S. Huang, A. Iurato, E. Jensen, J. Jin, U. Jørgensen, A. Kiesel, D.S. Kim, J. Liu, J.P. McCalmont, B.G. McMahon, M. Mos, P. Robson, E.J. Sacks, A. Sandu, G. Scalice, K. Schwarz, D. Scordia, R. Shafiei, I. Shield, G. Slavov, B.J. Stanton, K. Swaminathan, G. Taylor, A.F. Torres, L. M. Trindade, T. Tschaplinski, G.A. Tuskan, T. Yamada, C. Yeon Yu, R.S. Zalesny Jr., J. Zong, I. Lewandowski, Breeding progress and preparedness for mass-scale deployment of perennial lignocellulosic biomass crops switchgrass, miscanthus, willow and poplar, *Glob Change Biol Bioenergy* 11 (1) (2019) 118–151.
- [32] Á.L. Aradóttir, K. Svavarsdóttir, A.R. Baú, Clonal variability of native willows (*Salix pylicifolia* and *Salix lanata*) in Iceland and implications for use in restoration, *Icel. Agric. Sci.* 20 (2007) 61–72.
- [33] R.F. Kopp, L.P. Abrahamson, E.H. White, T.A. Volk, C.A. Nowak, R.C. Fillhart, Willow biomass production during ten successive annual harvests, *Biomass Bioenergy* 20 (1) (2001) 1–7.
- [34] M.J. Stolarski, S. Szczukowski, J. Tworowski, M. Krzyżaniak, D. Załuski, Willow biomass and cuttings' production potential over ten successive annual harvests, *Biomass Bioenergy* 105 (2017) 230–247.
- [35] E.S. Fabio, T.A. Volk, R.O. Miller, M.J. Serapiglia, H.G. Gauch, K.C.J. Van Rees, R. D. Hangs, B.Y. Amichev, Y.A. Kuzovkina, M. Labrecque, G.A. Johnson, R.G. Ewy, G.J. Kling, L.B. Smart, Genotype × environment interaction analysis of North American shrub willow yield trials confirms superior performance of triploid hybrids, *GCB Bioenergy* 9 (2) (2017) 445–459.
- [36] M.J. Aylott, E. Casella, I. Tubby, N.R. Street, P. Smith, G. Taylor, Yield and spatial supply of bioenergy poplar and willow short-rotation coppice in the UK, *New Phytol.* 178 (2) (2008) 358–370.
- [37] L.P. Abrahamson, T.A. Volk, R.F. Kopp, E.H. White, J.L. Ballard, *Willow Biomass Producers Handbook* (Revised), SUNY-ESF, Syracuse, N.Y., 2002.
- [38] R.C. Team, A Language and Environment for Statistical Computing, R Foundation for statistical computing, 2019. Vienna, Austria.
- [39] C.A. Souch, P.J. Martin, W. Stephens, G. Spoor, Effects of soil compaction and mechanical damage at harvest on growth and biomass production of short rotation coppice willow, *Plant Soil* 263 (1) (2004) 173–182.

- [40] M.J. Bullard, S.J. Mustill, P. Carver, P.M.I. Nixon, Yield improvements through modification of planting density and harvest frequency in short rotation coppice *Salix* spp.—2. Resource capture and use in two morphologically diverse varieties, *Biomass Bioenergy* 22 (1) (2002) 27–39.
- [41] H.G. Adegbi, T.A. Volk, E.H. White, L.P. Abrahamson, R.D. Briggs, D. H. Bickelhaupt, Biomass and nutrient removal by willow clones in experimental bioenergy plantations in New York State, *Biomass Bioenergy* 20 (6) (2001) 399–411.
- [42] Y. Von Fircks, L. Sennerby-Forsse, Seasonal fluctuations of starch in root and stem tissues of coppiced *Salix viminalis* plants grown under two nitrogen regimes, *Tree Physiol.* 18 (4) (1998) 243–249.
- [43] J. Moukoui, R.E. Farrell, K.J.C. Van Rees, R.K. Hynes, N. Bélanger, Intercropping *Caragana arborescens* with *Salix miyabeana* to satisfy nitrogen demand and maximize growth, *BioEnergy Research* 5 (3) (2012) 719–732.
- [44] H. Cochard, E. Casella, M. Mencuccini, Xylem vulnerability to cavitation varies among poplar and willow clones and correlates with yield, *Tree Physiol.* 27 (12) (2007) 1761–1767.
- [45] M.J. Bullard, S.J. Mustill, S.D. McMillan, P.M.I. Nixon, P. Carver, C.P. Britt, Yield improvements through modification of planting density and harvest frequency in short rotation coppice *Salix* spp.—1. Yield response in two morphologically diverse varieties, *Biomass Bioenergy* 22 (1) (2002) 15–25.